

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with @BENCH adt_fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT	SoA	Fold passes			Map passes		
	fields	bufs	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
OctTree	16	9	0.3759	0.4469	0.84×	0.3064	0.3574	0.86×
Compiler	9	8	0.1374	0.0665	2.07 ×	0.1595	0.1828	0.87×
DBQuery	15	13	0.1384	0.0981	1.41 ×	0.1377	0.1744	0.79×
DecisionTree	7	6	3.705	3.238	1.14 ×	—	—	—
DomTree	15	14	0.1516	0.0622	2.44 ×	0.0606	0.0805	0.75×
KDTree	17	16	0.1922	0.1416	1.36 ×	—	—	—
LinearListReduction	11	11	0.0305	0.0064	4.76 ×	—	—	—
List	2	2	0.1360	0.1114	1.22 ×	0.2587	0.3398	0.76×
MonoTree	3	2	0.0182	0.0227	0.80×	0.0955	0.0807	1.18 ×
ObjectGraph	5	4	0.0957	0.0937	1.02×	0.1422	0.1630	0.87×
PiecewiseFunctions	8	7	0.1414	0.1089	1.30 ×	0.2515	0.2819	0.89×
TernaryTree	5	3	0.0293	0.0270	1.09×	0.0833	0.0890	0.94×
Trie	10	9	0.0692	0.0456	1.52 ×	0.1525	0.1699	0.90×
ColorOctree	25	18	0.0897	0.0794	1.13 ×	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
OctTree	1.00×	0.75×	0.62×	0.46×	0.14×	0.61×	3.88 ×
Compiler	1.52 ×	0.52×	2.60 ×	0.84×	10.55 ×	0.34×	7.55 ×
DBQuery	1.07×	0.47×	1.29 ×	0.45×	8.05 ×	0.29×	5.80 ×
DecisionTree	1.19 ×	0.59×	16.95 ×	1.37 ×	0.90×	1.47 ×	34.53 ×
DomTree	1.65 ×	0.57×	2.04 ×	0.86×	16.74 ×	0.27×	10.01 ×
KDTree	1.38 ×	0.77×	1.20 ×	1.06×	0.20×	1.02×	4.80 ×
LinearListReduction	5.26 ×	0.90×	1.23 ×	2.12 ×	4.18 ×	3.35 ×	10.93 ×
List	0.84×	0.80×	3.97 ×	0.46×	0.14×	0.67×	1.47 ×
MonoTree	0.98×	0.85×	1.03×	0.65×	0.68×	0.73×	1.16 ×
ObjectGraph	0.97×	0.75×	3.05 ×	0.75×	0.35×	0.70×	2.87 ×
PiecewiseFunctions	1.10×	0.60×	2.38 ×	0.51×	1.84 ×	0.44×	4.50 ×
TernaryTree	1.15 ×	0.79×	0.32×	0.51×	0.19×	1.08×	1.47 ×
Trie	1.17 ×	0.56×	3.23 ×	0.83×	3.49 ×	0.22×	5.97 ×
Geomean	1.29 ×	0.67×	1.86 ×	0.75×	1.15 ×	0.64×	4.73 ×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
OctTree	2.6e+09 /2.6e+09	8.2e+09 /1.1e+10	1.5e+07 /2.3e+07	3.8e+05 /8.1e+05	7.1e+04 /4.9e+05	1.7e+04 /2.8e+04	6.5e+07/ 1.7e+07
Compiler	9.1e+08/ 6.0e+08	9.9e+08 /1.9e+09	1.3e+07/ 5.0e+06	2.8e+05 /3.3e+05	1.2e+06/ 1.2e+05	1.0e+04 /3.1e+04	5.1e+07/ 6.8e+06
DBQuery	9.5e+08/ 8.9e+08	1.1e+09 /2.2e+09	4.3e+06/ 3.3e+06	1.6e+05 /3.6e+05	8.3e+05/ 1.0e+05	8.2e+03 /2.8e+04	2.0e+07/ 3.5e+06
DecisionTree	1.9e+10/ 1.6e+10	4.0e+10 /6.8e+10	5.4e+08/ 3.2e+07	7.5e+04/ 5.5e+04	1.9e+06 /2.1e+06	7.5e+04/ 5.1e+04	1.1e+09/ 3.2e+07
DomTree	8.9e+08/ 5.4e+08	1.2e+09 /2.2e+09	1.2e+07/ 6.1e+06	1.0e+05 /1.2e+05	1.1e+06/ 6.4e+04	8.4e+03 /3.1e+04	5.7e+07/ 5.7e+06
KDTree	9.7e+08/ 7.1e+08	3.1e+09 /4.0e+09	1.4e+07/ 1.2e+07	1.9e+03/ 1.8e+03	2.3e+04 /1.1e+05	1.7e+03/ 1.6e+03	4.4e+07/ 9.2e+06
LinearListReduction	1.4e+08/ 2.6e+07	9.0e+07 /1.0e+08	1.5e+06/ 1.2e+06	6.8e+02/ 3.2e+02	2.8e+05/ 6.7e+04	5.9e+02/ 1.8e+02	1.2e+07/ 1.1e+06
List	1.3e+09 /1.5e+09	4.5e+09 /5.6e+09	9.4e+06/ 2.4e+06	2.6e+05 /5.5e+05	2.7e+04 /1.9e+05	7.6e+03 /1.1e+04	4.9e+07/ 3.3e+07
MonoTree	2.6e+08 /2.6e+08	9.1e+08 /1.1e+09	4.8e+05/ 4.7e+05	3.4e+04 /5.2e+04	8.5e+03 /1.3e+04	2.5e+03 /3.5e+03	2.1e+06/ 1.8e+06
ObjectGraph	8.6e+08 /8.9e+08	3.0e+09 /4.0e+09	9.8e+06/ 3.2e+06	1.8e+05 /2.4e+05	3.0e+04 /8.5e+04	1.2e+04 /1.8e+04	2.4e+07/ 8.4e+06
PiecewiseFunctions	1.3e+09/ 1.2e+09	2.8e+09 /4.7e+09	1.4e+07/ 6.0e+06	2.9e+05 /5.7e+05	2.0e+05/ 1.1e+05	2.5e+04 /5.7e+04	4.4e+07/ 9.8e+06
TernaryTree	4.2e+08/ 3.6e+08	1.2e+09 /1.5e+09	4.7e+05 /1.5e+06	6.2e+04 /1.2e+05	4.4e+03 /2.3e+04	2.6e+03/ 2.4e+03	3.4e+06/ 2.3e+06
Trie	6.8e+08/ 5.8e+08	1.2e+09 /2.2e+09	8.7e+06/ 2.7e+06	1.9e+05 /2.2e+05	1.7e+05/ 4.8e+04	8.0e+03 /3.6e+04	2.6e+07/ 4.3e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.2969	1.113	0.2493	<i>OOM</i>
DBQuery	0.2761	0.3419	0.2725	1.124
DecisionTree	3.705	5.685	3.238	25.12
DomTree	0.2122	0.3526	0.1427	0.5316
KDTree	0.1922	0.2592	0.1416	0.7330
LinearListReduction	0.0305	0.1356	0.0064	<i>OOM</i>
List	0.3947	<i>OOM</i>	0.4512	<i>OOM</i>
MonoTree	0.1137	0.1906	0.1034	0.2841
ObjectGraph	0.2379	0.5044	0.2566	0.7121
OctTree_sumMass	0.0349	0.0461	0.0322	<i>OOM</i>
OctTree_sumEnergy	0.0374	0.0611	0.0661	<i>OOM</i>
OctTree_countActive	0.0368	0.0481	0.0333	<i>OOM</i>
OctTree_countParticles	0.0377	0.0463	0.0375	<i>OOM</i>
OctTree_barnesHutPotential	0.0432	0.0631	0.0620	<i>OOM</i>
OctTree_fmmPotential	0.0963	0.1009	0.1364	<i>OOM</i>
OctTree_scaleEnergy	0.1507	0.1552	0.1793	<i>OOM</i>
OctTree_clearFlags	0.1557	0.1526	0.1781	<i>OOM</i>
PiecewiseFunctions	0.3930	0.5672	0.3908	2.155
TernaryTree	0.1125	0.1619	0.1160	0.6195
Trie	0.2218	0.3092	0.2155	1.008
ColorOctree	0.0897	0.1201	0.0794	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.19 $\times$	3.75 $\times$	4.46 $\times$	–
DBQuery	1.01 $\times$	1.24 $\times$	1.25 $\times$	0.30 $\times$
DecisionTree	1.14 $\times$	1.53 $\times$	1.76 $\times$	0.23 $\times$
DomTree	1.49 $\times$	1.66 $\times$	2.47 $\times$	0.66 $\times$
KDTree	1.36 $\times$	1.35 $\times$	1.83 $\times$	0.35 $\times$
LinearListReduction	4.76 $\times$	4.45 $\times$	21.16 $\times$	–
List	0.87 $\times$	–	–	–
MonoTree	1.10 $\times$	1.68 $\times$	1.84 $\times$	0.67 $\times$
ObjectGraph	0.93 $\times$	2.12 $\times$	1.97 $\times$	0.71 $\times$
OctTree_sumMass	1.08 $\times$	1.32 $\times$	1.43 $\times$	–
OctTree_sumEnergy	0.57 $\times$	1.64 $\times$	0.92 $\times$	–
OctTree_countActive	1.11 $\times$	1.31 $\times$	1.45 $\times$	–
OctTree_countParticles	1.01 $\times$	1.23 $\times$	1.24 $\times$	–
OctTree_barnesHutPotential	0.70 $\times$	1.46 $\times$	1.02 $\times$	–
OctTree_fmmPotential	0.71 $\times$	1.05 $\times$	0.74 $\times$	–
OctTree_scaleEnergy	0.84 $\times$	1.03 $\times$	0.87 $\times$	–
OctTree_clearFlags	0.87 $\times$	0.98 $\times$	0.86 $\times$	–
PiecewiseFunctions	1.01 $\times$	1.44 $\times$	1.45 $\times$	0.26 $\times$
TernaryTree	0.97 $\times$	1.44 $\times$	1.40 $\times$	0.26 $\times$
Trie	1.03 $\times$	1.39 $\times$	1.43 $\times$	0.31 $\times$
ColorOctree	1.13 $\times$	1.34 $\times$	1.51 $\times$	–

Table 6: Per-pass performance for `OctTree`, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
barnesHutPotential	F	11/16	31%	0.0432	0.0631	0.0620	–	0.70 $\times$	1.46 $\times$
1.02 $\times$	–								
clearFlags	M	15/16	6%	0.1557	0.1526	0.1781	–	0.87 $\times$	0.98 $\times$
0.86 $\times$	–								
countActive	F	10/16	38%	0.0368	0.0481	0.0333	–	1.11 $\times$	1.31 $\times$
1.45 $\times$	–								
countParticles	F	8/16	50%	0.0377	0.0463	0.0375	–	1.01 $\times$	1.23 $\times$
1.24 $\times$	–								
fmmPotential	F	12/16	25%	0.0963	0.1009	0.1364	–	0.71 $\times$	1.05 $\times$
0.74 $\times$	–								
scaleEnergy	M	16/16	0%	0.1507	0.1552	0.1793	–	0.84 $\times$	1.03 $\times$
0.87 $\times$	–								
sumEnergy	F	12/16	25%	0.0374	0.0611	0.0661	–	0.57 $\times$	1.64 $\times$
0.92 $\times$	–								
sumMass	F	10/16	38%	0.0349	0.0461	0.0322	–	1.08 $\times$	1.32 $\times$
1.43 $\times$	–								
paletteEntriesQuantized	F	13/25	48%	0.0438	0.0558	0.0366	–	1.20 $\times$	1.28 $\times$
1.52 $\times$	–								
quantizationErrorProxy	F	10/25	60%	0.0459	0.0643	0.0428	–	1.07 $\times$	1.40 $\times$
1.50 $\times$	–								
Total				0.6823	0.7937	0.8043	–	0.85 $\times$	1.16 $\times$
0.99 $\times$	–								
Geomean								0.89 $\times$	1.25 $\times$
1.12 $\times$	–								

Table 7: Per-pass PAPI counters for `OctTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
barnesHutPotential	F	2.2e+08/ 2.0e+08	8.2e+08 /1.0e+09	6.9e+04 /5.1e+05	4.4e+02 /5.6e+02	1.8e+03 /2.3e+04	3.1e+02 /4.0e+02	6.2e+06/ 1.8e+06
clearFlags	M	3.7e+08 /4.4e+08	8.0e+08 /1.7e+09	9.8e+05 /2.8e+06	2.2e+05 /4.1e+05	9.1e+03 /4.8e+04	7.4e+03 /1.2e+04	5.6e+06/ 2.4e+06
countActive	F	1.9e+08/ 1.6e+08	5.8e+08/ 5.8e+08	7.8e+05 /9.1e+05	6.3e+02/ 4.5e+02	1.3e+04 /2.3e+04	4.7e+02/ 2.7e+02	6.1e+06/ 3.0e+05
countParticles	F	1.9e+08/ 1.5e+08	5.7e+08/ 5.7e+08	9.3e+05/ 3.1e+05	6.9e+02/ 4.2e+02	3.9e+03 /7.9e+03	5.5e+02/ 2.0e+02	6.0e+06/ 1.1e+05
fmmPotential	F	4.9e+08 /5.1e+08	1.6e+09 /1.8e+09	2.5e+04 /7.9e+06	6.2e+02 /1.3e+03	1.8e+03 /1.5e+05	4.7e+02 /1.1e+03	4.5e+06/ 2.0e+06
scaleEnergy	M	3.5e+08 /4.4e+08	9.1e+08 /1.9e+09	1.4e+06 /2.7e+06	1.5e+05 /4.0e+05	8.4e+03 /4.3e+04	6.6e+03 /1.3e+04	5.6e+06/ 2.5e+06
sumEnergy	F	1.9e+08/ 1.7e+08	7.7e+08 /8.9e+08	3.4e+05 /2.8e+06	7.4e+02 /7.8e+02	2.2e+03 /6.1e+04	5.1e+02 /5.9e+02	6.3e+06/ 2.3e+06
sumMass	F	1.8e+08/ 1.6e+08	5.7e+08 /6.3e+08	8.0e+05 /2.4e+06	4.9e+02/ 4.2e+02	2.6e+03 /4.5e+04	3.8e+02/ 2.1e+02	6.2e+06/ 1.2e+06
paletteEntriesQuantized	F	2.1e+08/ 1.8e+08	6.7e+08 /7.6e+08	5.0e+06/ 2.1e+06	6.2e+02/ 5.1e+02	1.3e+04 /6.1e+04	4.6e+02/ 3.8e+02	9.3e+06/ 1.3e+06
quantizationErrorProxy	F	2.3e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.2e+06/ 9.5e+05	1.6e+02 /3.4e+02	1.6e+04 /3.4e+04	1.5e+02 /3.4e+02	9.3e+06/ 2.9e+06
Total		2.6e+09/2.6e+09	8.2e+09/1.1e+10	1.5e+07/2.3e+07	3.8e+05/8.1e+05	7.1e+04/4.9e+05	1.7e+04/2.8e+04	6.5e+07/1.7e+07

Table 8: Per-pass performance for `Compiler`, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0119	0.0797	0.0036	<i>OOM</i>	3.28 $\times$	6.71 $\times$	21.99 $\times$	–
branchStatsPass	F	2/9	78%	0.0115	0.0795	0.0045	<i>OOM</i>	2.54 $\times$	6.91 $\times$	17.54 $\times$	–
castInstCountPass	F	3/9	67%	0.0177	0.0804	0.0020	<i>OOM</i>	8.73 $\times$	4.55 $\times$	39.77 $\times$	–
has cycle	F	4/9	56%	0.0345	0.0934	0.0327	<i>OOM</i>	1.06 $\times$	2.71 $\times$	2.86 $\times$	–
instCountPass	F	2/9	78%	0.0125	0.0800	0.0042	<i>OOM</i>	2.95 $\times$	6.38 $\times$	18.84 $\times$	–
latencyModelPass	F	3/9	67%	0.0118	0.0801	0.0034	<i>OOM</i>	3.43 $\times$	6.80 $\times$	23.35 $\times$	–
memoryOpStatsPass	F	3/9	67%	0.0116	0.0794	0.0059	<i>OOM</i>	1.96 $\times$	6.87 $\times$	13.47 $\times$	–
stripSideEffectsPass	M	7/9	22%	0.0798	0.1870	0.0910	<i>OOM</i>	0.88 $\times$	2.34 $\times$	2.05 $\times$	–
targetReturnPass	M	9/9	0%	0.0797	0.2038	0.0918	<i>OOM</i>	0.87 $\times$	2.56 $\times$	2.22 $\times$	–
throughputModelPass	F	3/9	67%	0.0118	0.0798	0.0030	<i>OOM</i>	3.95 $\times$	6.78 $\times$	26.79 $\times$	–
verifyIR	F	9/9	0%	0.0142	0.0697	0.0071	<i>OOM</i>	2.01 $\times$	4.90 $\times$	9.85 $\times$	–
Total				0.2969	1.113	0.2493	–	1.19 $\times$	3.75 $\times$	4.46 $\times$	–
Geomean								2.29 $\times$	4.84 $\times$	11.09 $\times$	–

Table 9: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	5.9e+07/ 9.9e+06	5.1e+07/ 4.5e+07	1.2e+06/ 1.0e+05	2.7e+02/ 9.3e+01	2.2e+05/ 2.6e+03	2.5e+02/ 6.6e+01	4.8e+06/ 1.2e+04
branchStatsPass	F	5.8e+07/ 1.6e+07	7.9e+07 /8.6e+07	8.1e+05/ 2.3e+04	2.1e+02/ 1.1e+02	5.0e+04/ 3.4e+03	2.0e+02/ 9.3e+01	4.7e+06/ 4.8e+05
castInstCountPass	F	9.0e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.3e+04 /3.2e+04	1.5e+02/ 9.0e+01	8.3e+02 /1.1e+03	1.3e+02/ 6.9e+01	4.7e+06/ 1.8e+04
has cycle	F	1.3e+08/ 9.1e+07	1.1e+08 /1.5e+08	2.2e+06/ 1.1e+06	7.9e+04/ 5.9e+04	1.1e+05/ 2.2e+04	1.9e+03/ 6.5e+02	5.0e+06/ 1.4e+06
instCountPass	F	5.1e+07/ 1.1e+07	5.6e+07 /5.7e+07	1.3e+06/ 1.0e+05	2.8e+02/ 1.4e+02	2.5e+05/ 2.6e+03	2.5e+02/ 1.2e+02	4.8e+06/ 3.5e+04
latencyModelPass	F	6.0e+07/ 1.3e+07	6.2e+07 /6.9e+07	1.1e+06/ 3.5e+04	2.4e+02/ 8.8e+01	2.0e+05/ 4.0e+03	2.2e+02/ 6.5e+01	4.8e+06/ 6.2e+05
memoryOpStatsPass	F	5.9e+07/ 1.8e+07	8.5e+07 /9.1e+07	6.6e+05/ 3.1e+05	2.1e+02/ 1.1e+02	3.1e+04/ 1.2e+04	1.9e+02/ 9.9e+01	4.7e+06/ 4.5e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.9e+06/ 1.1e+06	1.0e+05 /1.4e+05	6.0e+03 /1.7e+04	2.8e+03 /8.4e+03	4.3e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /2.1e+08	1.7e+08 /6.3e+08	1.5e+06/ 1.3e+06	9.3e+04 /1.3e+05	5.0e+03 /2.2e+04	3.5e+03 /2.0e+04	3.8e+06/ 1.4e+06
throughputModelPass	F	6.0e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.1e+06/ 3.3e+04	2.8e+02/ 1.1e+02	2.0e+05/ 3.6e+03	2.3e+02/ 8.1e+01	4.8e+06/ 6.8e+05
verifyIR	F	4.5e+07/ 1.7e+07	7.4e+07 /8.6e+07	1.0e+06/ 7.9e+05	8.8e+02/ 6.9e+02	1.8e+05/ 2.8e+04	7.7e+02/ 5.6e+02	4.7e+06/ 4.2e+05
Total		9.1e+08/6.0e+08	9.9e+08/1.9e+09	1.3e+07/5.0e+06	2.8e+05/3.3e+05	1.2e+06/1.2e+05	1.0e+04/3.1e+04	5.1e+07/6.8e+06

Table 10: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.0e+08/−	1.6e+08/−	2.3e+06/−	4.9e+02/−	9.0e+04/−	4.6e+02/−	1.4e+07/−
branchStatsPass	F	4.0e+08/−	1.8e+08/−	2.1e+06/−	6.3e+02/−	4.3e+04/−	6.0e+02/−	1.5e+07/−
castInstCountPass	F	4.0e+08/−	1.8e+08/−	2.1e+06/−	6.2e+02/−	4.5e+04/−	5.8e+02/−	1.5e+07/−
has cycle	F	4.7e+08/−	2.1e+08/−	2.5e+06/−	5.3e+02/−	6.9e+04/−	5.1e+02/−	1.7e+07/−
instCountPass	F	4.0e+08/−	1.6e+08/−	2.2e+06/−	4.4e+02/−	9.8e+04/−	4.2e+02/−	1.4e+07/−
latencyModelPass	F	4.1e+08/−	1.7e+08/−	2.1e+06/−	6.0e+02/−	4.4e+04/−	5.9e+02/−	1.4e+07/−
memoryOpStatsPass	F	4.0e+08/−	1.9e+08/−	2.0e+06/−	3.4e+02/−	4.1e+04/−	3.3e+02/−	1.5e+07/−
stripSideEffectsPass	M	6.9e+08/−	3.3e+08/−	2.0e+06/−	9.9e+04/−	1.6e+04/−	2.0e+04/−	2.4e+07/−
targetReturnPass	M	6.6e+08/−	3.2e+08/−	2.2e+06/−	1.1e+05/−	1.8e+04/−	3.0e+04/−	2.2e+07/−
throughputModelPass	F	4.0e+08/−	1.7e+08/−	2.1e+06/−	8.1e+02/−	4.5e+04/−	7.3e+02/−	1.5e+07/−
verifyIR	F	3.5e+08/−	1.2e+08/−	2.4e+06/−	9.1e+02/−	2.0e+05/−	8.0e+02/−	1.1e+07/−
Total		5.0e+09/−	2.2e+09/−	2.4e+07/−	2.1e+05/−	7.1e+05/−	5.5e+04/−	1.8e+08/−

Table 11: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0686	0.0771	0.0907	0.3004	0.76 $\times$	1.12$\times$	0.85 $\times$	0.26 $\times$
countJoins	F	3/15	80%	0.0200	0.0285	0.0116	0.0850	1.73$\times$	1.42$\times$	2.46$\times$	0.33 $\times$
filterSelectivitySkew	F	5/15	67%	0.0198	0.0283	0.0123	0.0865	1.61$\times$	1.43$\times$	2.30$\times$	0.33 $\times$
hashJoinPressure	F	5/15	67%	0.0201	0.0326	0.0129	0.0863	1.56$\times$	1.62$\times$	2.53$\times$	0.38 $\times$
scaleCosts	M	15/15	0%	0.0691	0.0755	0.0837	0.3001	0.83 $\times$	1.09 $\times$	0.90 $\times$	0.25 $\times$
sumCost	F	6/15	60%	0.0309	0.0417	0.0232	0.0899	1.33$\times$	1.35$\times$	1.80$\times$	0.46 $\times$
sumMemory	F	6/15	60%	0.0196	0.0285	0.0150	0.0882	1.30$\times$	1.45$\times$	1.89$\times$	0.32 $\times$
sumRows	F	6/15	60%	0.0279	0.0297	0.0231	0.0872	1.21$\times$	1.06 $\times$	1.28$\times$	0.34 $\times$
Total				0.2761	0.3419	0.2725	1.124	1.01 $\times$	1.24$\times$	1.25$\times$	0.30 $\times$
Geomean								1.24$\times$	1.31$\times$	1.62$\times$	0.33 $\times$

Table 12: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.6e+08 /2.2e+08	1.9e+08 /7.2e+08	3.2e+05 /1.6e+06	8.7e+04 /1.8e+05	3.2e+04/ 2.8e+04	2.4e+03 /1.1e+04	2.3e+06/ 9.1e+05
countJoins	F	1.0e+08/ 5.8e+07	7.4e+07/ 7.1e+07	6.1e+05/ 2.5e+04	5.6e+02/ 2.4e+02	1.4e+05/ 2.1e+03	5.2e+02/ 1.6e+02	2.7e+06/ 5.7e+04
filterSelectivitySkew	F	1.0e+08/ 6.3e+07	1.0e+08 /1.1e+08	5.1e+05/ 7.2e+04	2.1e+02/ 1.4e+02	1.2e+05/ 6.2e+03	1.8e+02/ 1.2e+02	2.7e+06/ 3.2e+05
hashJoinPressure	F	1.0e+08/ 6.5e+07	8.4e+07 /9.9e+07	7.2e+05/ 1.0e+05	3.4e+02/ 2.1e+02	1.8e+05/ 7.4e+03	2.9e+02/ 1.9e+02	2.7e+06/ 1.4e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	3.2e+05 /1.2e+06	7.3e+04 /1.8e+05	3.5e+04/ 2.8e+04	3.6e+03 /1.7e+04	2.5e+06/ 1.1e+06
sumCost	F	8.7e+07/ 7.4e+07	9.0e+07 /1.2e+08	8.6e+05/ 7.7e+04	7.8e+02/ 3.9e+02	1.4e+05/ 1.1e+04	6.8e+02/ 2.8e+02	2.4e+06/ 3.5e+05
sumMemory	F	9.9e+07/ 7.6e+07	9.0e+07 /1.2e+08	6.7e+05/ 8.0e+04	3.1e+02/ 1.4e+02	1.6e+05/ 1.2e+04	2.8e+02/ 1.3e+02	2.7e+06/ 3.7e+05
sumRows	F	1.3e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.7e+05/ 1.1e+05	2.3e+02/ 2.0e+02	2.2e+04/ 7.6e+03	2.0e+02/ 1.9e+02	2.4e+06/ 3.1e+05
Total		9.5e+08/8.9e+08	1.1e+09/2.2e+09	4.3e+06/3.3e+06	1.6e+05/3.6e+05	8.3e+05/1.0e+05	8.2e+03/2.8e+04	2.0e+07/3.5e+06

Table 13: Per-pass PAPI counters for **DBQuery** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	6.6e+05 /2.0e+07	9.3e+04 /6.6e+05	1.5e+04 /4.6e+04	3.6e+03 /3.5e+04	2.1e+06/ 1.0e+06
countJoins	F	1.4e+08 /4.3e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.1e+05	2.7e+02 /7.8e+02	1.1e+05/ 1.3e+04	2.3e+02/ 2.0e+02	2.5e+06/ 8.8e+04
filterSelectivitySkew	F	1.4e+08 /4.4e+08	2.0e+08 /7.9e+08	1.5e+06/ 8.3e+05	2.1e+02 /1.3e+03	6.9e+04/ 1.7e+04	2.0e+02 /4.8e+02	2.4e+06/ 2.3e+05
hashJoinPressure	F	1.6e+08 /4.4e+08	1.8e+08 /7.7e+08	1.7e+06/ 6.4e+05	2.5e+02 /9.7e+02	1.1e+05/ 1.7e+04	2.2e+02 /3.7e+02	2.3e+06/ 1.6e+05
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.7e+05 /2.3e+07	7.3e+04 /3.1e+05	1.7e+04 /5.9e+04	2.2e+03 /5.1e+04	2.3e+06/ 1.2e+06
sumCost	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	2.4e+06/ 1.7e+06	5.2e+02 /1.6e+03	1.3e+05/ 2.8e+04	4.2e+02 /8.8e+02	2.2e+06/ 3.0e+05
sumMemory	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.1e+06	4.1e+02 /1.1e+03	7.3e+04/ 2.1e+04	4.0e+02 /6.1e+02	2.4e+06/ 3.3e+05
sumRows	F	1.5e+08 /4.4e+08	2.3e+08 /8.3e+08	1.6e+06/ 1.2e+06	2.5e+02 /1.2e+03	6.2e+04/ 2.2e+04	2.5e+02 /6.8e+02	2.3e+06/ 3.5e+05
Total		1.3e+09/5.3e+09	1.8e+09/9.5e+09	1.2e+07/4.8e+07	1.7e+05/9.8e+05	5.9e+05/2.2e+05	7.5e+03/8.9e+04	1.8e+07/3.6e+06

Table 14: Per-pass performance for `DecisionTree`, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.179	4.308	2.885	20.99	1.10 $\times$	1.35 $\times$	1.49 $\times$	0.21 $\times$
classify Depth	F	4/7	43%	0.0454	0.0613	0.0389	0.3037	1.17 $\times$	1.35 $\times$	1.57 $\times$	0.20 $\times$
classify tree	F	5/7	29%	0.0453	0.0613	0.0393	0.2980	1.15 $\times$	1.35 $\times$	1.56 $\times$	0.21 $\times$
countFeatureUses	F	3/7	57%	0.0486	0.1416	0.0314	0.3922	1.55 $\times$	2.92 $\times$	4.51 $\times$	0.36 $\times$
countLeaves	F	2/7	71%	0.0462	0.1415	0.0262	0.3825	1.77 $\times$	3.06 $\times$	5.40 $\times$	0.37 $\times$
countNodes	F	2/7	71%	0.0460	0.1417	0.0286	0.4108	1.61 $\times$	3.08 $\times$	4.96 $\times$	0.34 $\times$
countSmallLeaves	F	3/7	57%	0.0491	0.1368	0.0321	0.3833	1.53 $\times$	2.79 $\times$	4.27 $\times$	0.36 $\times$
inferenceCost	F	2/7	71%	0.0492	0.1396	0.0290	0.3831	1.70 $\times$	2.84 $\times$	4.81 $\times$	0.36 $\times$
sumImpurity	F	3/7	57%	0.0486	0.1389	0.0327	0.3951	1.49 $\times$	2.86 $\times$	4.25 $\times$	0.35 $\times$
sumPathLengths	F	3/7	57%	0.0512	0.1398	0.0336	0.4034	1.52 $\times$	2.73 $\times$	4.16 $\times$	0.35 $\times$
sumSamples	F	3/7	57%	0.0465	0.1354	0.0326	0.3993	1.43 $\times$	2.91 $\times$	4.16 $\times$	0.34 $\times$
treeDepth	F	2/7	71%	0.0496	0.1391	0.0286	0.3829	1.73 $\times$	2.81 $\times$	4.87 $\times$	0.36 $\times$
Total				3.705	5.685	3.238	25.12	1.14 $\times$	1.53 $\times$	1.76 $\times$	0.23 $\times$
Geomean								1.46 $\times$	2.39 $\times$	3.49 $\times$	0.31 $\times$

Table 15: Per-pass PAPI counters for `DecisionTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.6e+10/ 1.4e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.7e+07	6.4e+04/ 4.8e+04	1.6e+06 /2.0e+06	6.4e+04/ 4.5e+04	9.3e+08/ 1.9e+07
classify Depth	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.6e+06/ 4.1e+05	9.0e+02/ 3.1e+02	2.1e+04 /2.8e+04	8.8e+02/ 2.8e+02	1.3e+07/ 3.3e+05
classify tree	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.6e+06/ 4.1e+05	1.1e+03/ 3.6e+02	2.0e+04 /2.8e+04	1.1e+03/ 3.1e+02	1.3e+07/ 3.4e+05
countFeatureUses	F	2.5e+08/ 1.6e+08	7.5e+08 /7.8e+08	7.7e+06/ 3.9e+05	7.2e+02/ 5.0e+02	3.5e+04/ 1.4e+04	6.8e+02/ 4.8e+02	1.5e+07/ 2.1e+06
countLeaves	F	2.3e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.4e+06/ 7.6e+05	1.3e+03/ 8.0e+02	4.5e+04/ 1.7e+04	1.2e+03/ 7.2e+02	1.4e+07/ 2.3e+05
countNodes	F	2.3e+08/ 1.4e+08	6.1e+08 /6.2e+08	7.3e+06/ 7.6e+05	1.5e+03/ 8.4e+02	5.3e+04/ 1.7e+04	1.3e+03/ 7.3e+02	1.4e+07/ 2.3e+05
countSmallLeaves	F	2.5e+08/ 1.6e+08	7.1e+08 /8.4e+08	5.9e+06/ 9.0e+04	7.5e+02/ 6.7e+02	3.2e+04/ 1.3e+04	7.4e+02/ 6.3e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.2e+05	1.1e+03/ 5.8e+02	1.7e+04/ 7.9e+03	1.1e+03/ 5.4e+02	1.4e+07/ 4.1e+05
sumImpurity	F	2.5e+08/ 1.7e+08	7.6e+08 /7.8e+08	7.8e+06/ 1.1e+06	1.4e+03/ 5.7e+02	3.6e+04/ 1.4e+04	1.4e+03/ 5.4e+02	1.5e+07/ 2.1e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.7e+06/ 4.8e+05	1.1e+03/ 5.8e+02	3.9e+04/ 1.4e+04	1.0e+03/ 5.6e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.4e+08/ 1.7e+08	6.4e+08 /7.7e+08	6.2e+06/ 2.3e+05	7.5e+02/ 7.4e+02	4.1e+04/ 1.4e+04	7.3e+02/ 7.2e+02	1.5e+07/ 2.1e+06
treeDepth	F	2.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.2e+06/ 7.4e+05	1.1e+03/ 6.7e+02	1.9e+04/ 1.6e+04	1.1e+03/ 6.3e+02	1.4e+07/ 2.3e+05
Total		1.9e+10/1.6e+10	4.0e+10/6.8e+10	5.4e+08/3.2e+07	7.5e+04/5.5e+04	1.9e+06/2.1e+06	7.5e+04/5.1e+04	1.1e+09/3.2e+07

Table 16: Per-pass PAPI counters for `DecisionTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.2e+10 /1.1e+11	5.1e+10 /1.9e+11	1.5e+07 /4.8e+07	2.0e+04 /1.3e+05	3.8e+05 /2.2e+06	2.0e+04/ 1.8e+04	8.6e+08/ 2.0e+07
classify Depth	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.6e+05 /6.6e+05	3.1e+02 /1.8e+03	4.9e+03 /3.1e+04	2.9e+02 /3.8e+02	1.2e+07/ 3.3e+05
classify tree	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.6e+05 /6.8e+05	2.5e+02 /2.1e+03	4.8e+03 /3.2e+04	2.4e+02 /4.1e+02	1.2e+07/ 3.3e+05
countFeatureUses	F	7.2e+08 /2.0e+09	1.3e+09 /3.1e+09	2.0e+06 /3.2e+06	2.8e+02 /2.1e+03	9.3e+03 /7.0e+04	2.5e+02 /9.2e+02	9.4e+06/ 1.2e+06
countLeaves	F	7.1e+08 /1.9e+09	1.2e+09 /2.9e+09	4.3e+06/ 7.2e+05	7.3e+02 /3.3e+03	1.3e+04 /3.3e+04	6.6e+02/ 5.6e+02	9.2e+06/ 3.6e+05
countNodes	F	7.1e+08 /1.9e+09	1.2e+09 /3.0e+09	4.5e+06/ 8.3e+05	9.2e+02 /2.8e+03	1.8e+04 /3.5e+04	7.8e+02/ 5.8e+02	9.2e+06/ 3.1e+05
countSmallLeaves	F	6.9e+08 /1.9e+09	1.3e+09 /2.9e+09	1.1e+06 /3.4e+06	4.5e+02 /2.2e+03	6.8e+03 /7.3e+04	4.1e+02 /8.5e+02	9.3e+06/ 1.2e+06
inferenceCost	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	7.2e+02 /1.6e+03	1.4e+04 /3.1e+04	6.9e+02/ 3.9e+02	9.2e+06/ 3.8e+05
sumImpurity	F	7.0e+08 /2.0e+09	1.3e+09 /3.0e+09	2.2e+06 /3.5e+06	3.9e+02 /5.1e+02	1.0e+04 /7.5e+04	3.6e+02 /4.5e+02	9.5e+06/ 1.2e+06
sumPathLengths	F	7.1e+08 /2.0e+09	1.3e+09 /3.1e+09	1.5e+06 /3.6e+06	6.2e+02 /3.6e+03	8.1e+03 /7.9e+04	6.1e+02 /1.1e+03	9.2e+06/ 1.2e+06
sumSamples	F	6.8e+08 /2.0e+09	1.2e+09 /3.0e+09	1.1e+06 /3.6e+06	9.0e+02 /2.6e+03	6.9e+03 /7.7e+04	8.7e+02 /1.0e+03	9.3e+06/ 1.2e+06
treeDepth	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.3e+05	4.3e+02 /1.6e+03	1.4e+04 /3.1e+04	4.0e+02/ 3.3e+02	9.2e+06/ 4.0e+05
Total		2.9e+10/1.3e+11	6.4e+10/2.2e+11	4.1e+07/6.9e+07	2.6e+04/1.5e+05	4.9e+05/2.8e+06	2.5e+04/2.5e+04	9.7e+08/2.8e+07

Table 17: Per-pass performance for `DomTree`, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0393	0.0787	0.0212	0.0915	1.85 $\times$	2.00 $\times$	3.71 $\times$	0.86 $\times$
computeWidths	M	14/15	7%	0.0310	0.0315	0.0443	0.0882	0.70 $\times$	1.01 $\times$	0.71 $\times$	0.36 $\times$
count styled	F	3/15	80%	0.0375	0.0704	0.0117	0.0880	3.21 $\times$	1.88 $\times$	6.03 $\times$	0.80 $\times$
find max Bottom	F	5/15	67%	0.0373	0.0702	0.0183	0.0881	2.05 $\times$	1.88 $\times$	3.84 $\times$	0.80 $\times$
scaleLayout	M	15/15	0%	0.0296	0.0308	0.0363	0.0872	0.82 $\times$	1.04 $\times$	0.85 $\times$	0.35 $\times$
sumTextWidth	F	3/15	80%	0.0375	0.0711	0.0111	0.0886	3.38 $\times$	1.90 $\times$	6.42 $\times$	0.80 $\times$
Total				0.2122	0.3526	0.1427	0.5316	1.49 $\times$	1.66 $\times$	2.47 $\times$	0.66 $\times$
Geomean								1.69 $\times$	1.56 $\times$	2.63 $\times$	0.62 $\times$

Table 18: Per-pass PAPI counters for `DomTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	2.0e+08/ 9.8e+07	2.9e+08 /4.9e+08	2.7e+06/ 1.3e+06	1.0e+03/ 4.1e+02	1.6e+05/ 2.4e+04	8.1e+02/ 3.1e+02	1.3e+07/ 1.8e+06
computeWidths	M	6.6e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.6e+06	5.3e+04 /6.9e+04	4.6e+03 /1.0e+04	2.3e+03 /1.5e+04	1.3e+06/ 4.5e+05
count styled	F	1.9e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.3e+06/ 2.5e+04	1.1e+03/ 1.5e+02	2.9e+05/ 4.9e+03	1.0e+03/ 1.4e+02	1.4e+07/ 7.2e+05
find max Bottom	F	1.9e+08/ 9.3e+07	2.5e+08 /4.1e+08	2.6e+06/ 8.9e+05	3.8e+02/ 1.1e+02	1.6e+05/ 1.2e+04	3.5e+02/ 9.4e+01	1.4e+07/ 1.6e+06
scaleLayout	M	6.0e+07 /9.9e+07	8.8e+07 /2.9e+08	1.1e+06 /2.3e+06	4.7e+04 /5.0e+04	2.4e+04/ 9.2e+03	2.5e+03 /1.5e+04	1.4e+06/ 4.4e+05
sumTextWidth	F	1.9e+08/ 5.6e+07	2.0e+08 /2.4e+08	2.6e+06/ 2.0e+04	1.5e+03/ 1.6e+02	4.4e+05/ 4.3e+03	1.4e+03/ 1.5e+02	1.3e+07/ 7.3e+05
Total		8.9e+08/5.4e+08	1.2e+09/2.2e+09	1.2e+07/6.1e+06	1.0e+05/1.2e+05	1.1e+06/6.4e+04	8.4e+03/3.1e+04	5.7e+07/5.7e+06

Table 19: Per-pass PAPI counters for `DomTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.9e+08 /4.6e+08	4.4e+08 /1.8e+09	1.1e+07/ 2.2e+06	6.1e+02 /1.5e+03	2.3e+05/ 3.6e+04	4.5e+02 /1.3e+03	9.6e+06/ 1.4e+06
computeWidths	M	6.9e+07 /3.6e+08	1.5e+08 /9.0e+08	1.3e+06 /3.6e+06	4.7e+04 /2.4e+05	4.2e+03 /1.4e+04	1.8e+03 /2.3e+04	1.2e+06/ 4.8e+05
count styled	F	3.6e+08 /4.5e+08	4.3e+08 /1.7e+09	1.0e+07/ 1.2e+06	2.5e+02 /4.7e+02	3.9e+04/ 3.7e+04	2.2e+02 /4.0e+02	1.1e+07/ 4.7e+05
find max Bottom	F	3.6e+08 /4.5e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.5e+06	2.4e+02 /9.9e+02	1.8e+04 /2.9e+04	2.1e+02 /9.5e+02	1.0e+07/ 1.0e+06
scaleLayout	M	6.8e+07 /3.5e+08	1.1e+08 /7.6e+08	9.8e+05 /4.1e+06	5.1e+04 /8.2e+04	1.1e+04 /1.2e+04	1.4e+03 /2.3e+04	1.3e+06/ 4.7e+05
sumTextWidth	F	3.6e+08 /4.5e+08	3.9e+08 /1.7e+09	1.0e+07/ 1.3e+06	3.0e+02 /1.3e+03	8.8e+04/ 4.0e+04	2.8e+02 /9.3e+02	1.0e+07/ 4.5e+05
Total		1.6e+09/2.5e+09	2.0e+09/8.6e+09	4.3e+07/1.4e+07	1.0e+05/3.3e+05	3.9e+05/1.7e+05	4.4e+03/5.0e+04	4.3e+07/4.3e+06

Table 20: Per-pass performance for `KDTree`, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0303	0.0463	0.0236	0.1220	1.28 $\times$	1.53 $\times$	1.96 $\times$	0.38 $\times$
countInRange tight_box	F	13/17	24%	0.0300	0.0415	0.0198	0.1265	1.51 $\times$	1.39 $\times$	2.10 $\times$	0.33 $\times$
photonMappingBenchmark	F	12/17	29%	0.0447	0.0491	0.0410	0.1276	1.09 $\times$	1.10 $\times$	1.20 $\times$	0.39 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0270	0.0401	0.0164	0.1158	1.64 $\times$	1.49 $\times$	2.44 $\times$	0.35 $\times$
sumMassInRange	F	14/17	18%	0.0294	0.0415	0.0201	0.1260	1.46 $\times$	1.41 $\times$	2.07 $\times$	0.33 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0309	0.0406	0.0206	0.1151	1.50 $\times$	1.31 $\times$	1.97 $\times$	0.35 $\times$
Total				0.1922	0.2592	0.1416	0.7330	1.36 $\times$	1.35 $\times$	1.83 $\times$	0.35 $\times$
Geomean								1.40 $\times$	1.36 $\times$	1.91 $\times$	0.35 $\times$

Table 21: Per-pass PAPI counters for `KDTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.5e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.7e+06/ 2.2e+06	5.4e+02/ 4.8e+02	4.4e+03 /3.1e+04	4.2e+02/ 3.6e+02	7.7e+06/ 1.5e+06
countInRange tight_box	F	1.5e+08/ 1.0e+08	4.0e+08 /5.5e+08	2.2e+06/ 1.9e+06	2.2e+02/ 1.8e+02	3.8e+03 /1.7e+04	1.9e+02/ 1.5e+02	7.7e+06/ 1.7e+06
photonMappingBenchmark	F	2.3e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.5e+06	6.6e+02/ 6.1e+02	4.1e+03 /1.5e+04	6.2e+02/ 5.9e+02	5.7e+06/ 1.5e+06
pointCloudNeighborhood	F	1.4e+08/ 8.3e+07	3.3e+08 /4.8e+08	2.6e+06/ 1.4e+06	1.5e+02/ 1.5e+02	4.0e+03 /1.6e+04	1.4e+02/ 1.3e+02	7.7e+06/ 1.4e+06
sumMassInRange	F	1.5e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.4e+06 /3.0e+06	1.9e+02 /2.1e+02	3.6e+03 /2.0e+04	1.5e+02 /2.1e+02	7.6e+06/ 1.8e+06
twoPointCorrelation bin_8_16	F	1.6e+08/ 1.0e+08	4.8e+08 /6.1e+08	2.1e+06/ 6.8e+05	1.5e+02 /1.9e+02	3.0e+03 /1.3e+04	1.3e+02 /1.7e+02	7.7e+06/ 1.3e+06
Total		9.7e+08/7.1e+08	3.1e+09/4.0e+09	1.4e+07/1.2e+07	1.9e+03/1.8e+03	2.3e+04/1.1e+05	1.7e+03/1.6e+03	4.4e+07/9.2e+06

Table 22: Per-pass PAPI counters for `KDTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.1e+08 /6.1e+08	4.9e+08 /1.3e+09	5.2e+06 /5.5e+06	5.2e+02 /2.3e+03	5.6e+04/ 3.7e+04	3.7e+02 /2.2e+03	5.5e+06/ 1.5e+06
countInRange tight_box	F	2.1e+08 /6.4e+08	4.5e+08 /1.3e+09	5.1e+06 /5.8e+06	2.1e+02 /2.5e+03	5.0e+03 /2.2e+04	1.9e+02 /2.4e+03	5.8e+06/ 1.6e+06
photonMappingBenchmark	F	2.5e+08 /6.5e+08	1.1e+09 /2.0e+09	2.8e+06 /5.9e+06	5.5e+02 /2.1e+03	3.7e+03 /2.6e+04	5.2e+02 /2.1e+03	5.2e+06/ 1.5e+06
pointCloudNeighborhood	F	2.0e+08 /5.9e+08	4.1e+08 /1.3e+09	5.0e+06/ 4.8e+06	2.9e+02 /1.3e+03	6.4e+03 /2.1e+04	2.7e+02 /1.3e+03	5.9e+06/ 1.4e+06
sumMassInRange	F	2.1e+08 /6.4e+08	4.7e+08 /1.3e+09	5.0e+06 /6.3e+06	2.7e+02 /1.6e+03	4.6e+03 /2.6e+04	2.5e+02 /1.6e+03	5.7e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.1e+08 /5.8e+08	6.1e+08 /1.4e+09	4.7e+06/ 3.7e+06	2.8e+02 /1.5e+03	7.5e+03 /1.8e+04	2.7e+02 /1.5e+03	5.9e+06/ 1.3e+06
Total		1.3e+09/3.7e+09	3.5e+09/8.4e+09	2.8e+07/3.2e+07	2.1e+03/1.1e+04	8.3e+04/1.5e+05	1.9e+03/1.1e+04	3.4e+07/8.9e+06

Table 23: Per-pass performance for `LinearListReduction`, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0305	0.1356	0.0064	<i>OOM</i>	4.76 $\times$	4.45 $\times$	21.16 $\times$	–
Total				0.0305	0.1356	0.0064	–	4.76 $\times$	4.45 $\times$	21.16 $\times$	–
Geomean								4.76 $\times$	4.45 $\times$	21.16 $\times$	–

Table 24: Per-pass PAPI counters for `LinearListReduction` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.4e+08/ 2.6e+07	9.0e+07 /1.0e+08	1.5e+06/ 1.2e+06	6.8e+02/ 3.2e+02	2.8e+05/ 6.7e+04	5.9e+02/ 1.8e+02	1.2e+07/ 1.1e+06
Total		1.4e+08/2.6e+07	9.0e+07/1.0e+08	1.5e+06/1.2e+06	6.8e+02/3.2e+02	2.8e+05/6.7e+04	5.9e+02/1.8e+02	1.2e+07/1.1e+06

Table 25: Per-pass PAPI counters for `LinearListReduction` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	6.8e+08/–	2.5e+08/–	4.0e+06/–	1.3e+03/–	3.9e+05/–	1.1e+03/–	2.7e+07/–
Total		6.8e+08/–	2.5e+08/–	4.0e+06/–	1.3e+03/–	3.9e+05/–	1.1e+03/–	2.7e+07/–

Table 26: Per-pass performance for `List`, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2587	<i>OOM</i>	0.3398	<i>OOM</i>	0.76 $\times$	–	–	–
length List	F	1/2	50%	0.0456	<i>OOM</i>	0.0216	<i>OOM</i>	2.11 $\times$	–	–	–
sumList	F	2/2	0%	0.0451	<i>OOM</i>	0.0453	<i>OOM</i>	1.00 $\times$	–	–	–
sumList tail recursive	F	2/2	0%	0.0453	<i>OOM</i>	0.0445	<i>OOM</i>	1.02 $\times$	–	–	–
Total				0.3947	–	0.4512	–	0.87 $\times$	–	–	–
Geomean								1.13 $\times$	–	–	–

Table 27: Per-pass PAPI counters for `List` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.7e+08	1.9e+09 /2.8e+09	8.1e+06/ 1.1e+06	2.6e+05 /5.5e+05	1.7e+04 /5.3e+04	7.2e+03 /1.1e+04	9.8e+06/ 7.8e+06
length List	F	2.3e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.2e+05/ 1.1e+05	1.9e+02/ 1.2e+02	3.0e+03 /1.4e+04	1.8e+02/ 1.0e+02	1.3e+07/ 1.2e+06
sumList	F	2.3e+08 /2.3e+08	9.0e+08 /1.0e+09	6.4e+05/ 5.4e+05	1.2e+02 /1.3e+02	3.3e+03 /6.8e+04	1.1e+02 /1.2e+02	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.3e+08/ 2.3e+08	9.0e+08 /1.0e+09	5.2e+05 /5.7e+05	1.9e+02 /2.0e+02	3.2e+03 /5.8e+04	1.6e+02 /1.9e+02	1.3e+07/ 1.2e+07
Total		1.3e+09/1.5e+09	4.5e+09/5.6e+09	9.4e+06/2.4e+06	2.6e+05/5.5e+05	2.7e+04/1.9e+05	7.6e+03/1.1e+04	4.9e+07/3.3e+07

Table 28: Per-pass PAPI counters for `List` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 29: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0955	0.1102	0.0807	0.1221	1.18 $\times$	1.15 $\times$	1.37 $\times$	0.90 $\times$
sumTree	F	3/3	0%	0.0094	0.0403	0.0115	0.0809	0.82 $\times$	4.31 $\times$	3.51 $\times$	0.50 $\times$
sumTree TailRec	F	3/3	0%	0.0088	0.0401	0.0112	0.0812	0.79 $\times$	4.54 $\times$	3.57 $\times$	0.49 $\times$
Total				0.1137	0.1906	0.1034	0.2841	1.10 $\times$	1.68 $\times$	1.84 $\times$	0.67 $\times$
Geomean								0.91 $\times$	2.83 $\times$	2.58 $\times$	0.61 $\times$

Table 30: Per-pass PAPI counters for **MonoTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	4.7e+05/ 4.2e+05	3.3e+04 /5.1e+04	7.0e+03/ 4.8e+03	2.1e+03 /3.0e+03	4.4e+05 /4.4e+05
sumTree	F	4.7e+07 /5.8e+07	2.2e+08 /2.6e+08	8.2e+03 /3.5e+04	2.8e+02 /2.9e+02	7.5e+02 /3.8e+03	2.5e+02/ 2.5e+02	8.4e+05/ 6.9e+05
sumTree TailRec	F	4.4e+07 /5.7e+07	1.9e+08 /2.2e+08	6.5e+03 /2.1e+04	2.6e+02 /3.2e+02	7.5e+02 /3.9e+03	2.3e+02 /2.8e+02	8.3e+05/ 6.9e+05
Total		2.6e+08/2.6e+08	9.1e+08/1.1e+09	4.8e+05/4.7e+05	3.4e+04/5.2e+04	8.5e+03/1.3e+04	2.5e+03/3.5e+03	2.1e+06/1.8e+06

Table 31: Per-pass PAPI counters for **MonoTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.1e+08	7.1e+08 /1.2e+09	2.8e+05 /1.1e+06	3.4e+04 /5.3e+04	4.3e+03 /1.6e+04	1.8e+03 /2.8e+03	3.7e+05 /4.2e+05
sumTree	F	2.0e+08 /4.1e+08	4.3e+08 /6.3e+08	7.5e+04 /9.4e+05	4.2e+02/ 3.8e+02	1.1e+03 /1.5e+04	3.5e+02/ 3.3e+02	3.7e+05 /4.1e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	6.8e+04 /9.3e+05	3.7e+02/ 2.6e+02	1.2e+03 /1.4e+04	3.2e+02/ 2.0e+02	3.7e+05 /4.1e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	4.3e+05/2.9e+06	3.5e+04/5.4e+04	6.6e+03/4.6e+04	2.5e+03/3.3e+03	1.1e+06/1.2e+06

Table 32: Per-pass performance for **ObjectGraph**, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0134	0.0446	0.0107	0.0464	1.25 $\times$	3.34 $\times$	4.18 $\times$	0.96 $\times$
countMarked	F	3/5	40%	0.0134	0.0446	0.0139	0.0468	0.96 $\times$	3.32 $\times$	3.20 $\times$	0.95 $\times$
countSurvivors	F	4/5	20%	0.0143	0.0449	0.0132	0.0469	1.08 $\times$	3.15 $\times$	3.40 $\times$	0.96 $\times$
deadBytes	F	4/5	20%	0.0141	0.0440	0.0126	0.0471	1.11 $\times$	3.13 $\times$	3.48 $\times$	0.93 $\times$
liveBytes	F	4/5	20%	0.0139	0.0433	0.0127	0.0471	1.10 $\times$	3.11 $\times$	3.42 $\times$	0.92 $\times$
sumObjIds	F	3/5	40%	0.0133	0.0433	0.0105	0.0471	1.27 $\times$	3.25 $\times$	4.12 $\times$	0.92 $\times$
sweepUnmarked	M	5/5	0%	0.0717	0.0899	0.0814	0.1888	0.88 $\times$	1.26 $\times$	1.10 $\times$	0.48 $\times$
totalHeapSize	F	3/5	40%	0.0134	0.0615	0.0200	0.0501	0.67 $\times$	4.61 $\times$	3.08 $\times$	1.23 $\times$
touchHotObjects	M	5/5	0%	0.0706	0.0882	0.0815	0.1918	0.87 $\times$	1.25 $\times$	1.08 $\times$	0.46 $\times$
Total				0.2379	0.5044	0.2566	0.7121	0.93 $\times$	2.12 $\times$	1.97 $\times$	0.71 $\times$
Geomean								1.00 $\times$	2.71 $\times$	2.72 $\times$	0.83 $\times$

Table 33: Per-pass PAPI counters for **ObjectGraph** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.8e+07/ 5.4e+07	2.6e+08 /2.8e+08	1.2e+06/ 4.7e+04	5.7e+02/ 2.9e+02	3.6e+03 /3.8e+03	5.4e+02/ 2.6e+02	3.0e+06/ 7.1e+05
countMarked	F	6.8e+07/ 5.7e+07	2.6e+08 /2.8e+08	1.2e+06/ 4.4e+05	4.1e+02 /4.2e+02	3.7e+03 /1.1e+04	3.9e+02/ 3.7e+02	3.0e+06/ 6.5e+05
countSurvivors	F	7.2e+07/ 6.7e+07	3.1e+08 /3.8e+08	1.0e+06/ 7.2e+04	3.1e+02 /3.4e+02	2.3e+03 /5.4e+03	2.8e+02 /3.3e+02	2.9e+06/ 1.3e+06
deadBytes	F	7.1e+07/ 6.4e+07	2.7e+08 /3.5e+08	7.6e+05/ 6.5e+04	3.3e+02 /3.9e+02	2.2e+03 /5.8e+03	3.3e+02 /3.6e+02	2.8e+06/ 1.3e+06
liveBytes	F	7.0e+07/ 6.4e+07	2.7e+08 /3.5e+08	8.9e+05/ 6.1e+04	4.0e+02 /4.1e+02	2.4e+03 /5.8e+03	3.8e+02 /4.0e+02	2.9e+06/ 1.3e+06
sumObjIds	F	6.8e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.2e+06/ 1.0e+05	2.0e+02 /3.2e+02	2.8e+03 /4.5e+03	1.8e+02 /2.8e+02	2.9e+06/ 6.9e+05
sweepUnmarked	M	1.9e+08 /2.4e+08	5.5e+08 /8.8e+08	1.1e+06/ 5.8e+05	7.2e+04 /1.1e+05	5.9e+03 /1.1e+04	5.7e+03 /8.3e+03	2.0e+06/ 1.0e+06
totalHeapSize	F	6.7e+07/ 5.8e+07	2.5e+08 /2.7e+08	1.2e+06 /1.3e+06	7.9e+02/ 6.6e+02	2.7e+03 /2.5e+04	6.4e+02/ 5.7e+02	2.9e+06/ 4.9e+05
touchHotObjects	M	1.9e+08 /2.3e+08	5.7e+08 /9.0e+08	1.2e+06/ 5.7e+05	1.0e+05 /1.3e+05	4.4e+03 /1.2e+04	4.0e+03 /6.8e+03	1.8e+06/ 1.0e+06
Total		8.6e+08/8.9e+08	3.0e+09/4.0e+09	9.8e+06/3.2e+06	1.8e+05/2.4e+05	3.0e+04/8.5e+04	1.2e+04/1.8e+04	2.4e+07/8.4e+06

Table 34: Per-pass PAPI counters for **ObjectGraph** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.4e+05/ 4.6e+05	4.0e+02/ 2.2e+02	4.5e+03 /8.2e+03	3.2e+02/ 1.9e+02	1.8e+06/ 4.2e+05
countMarked	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.3e+05/ 4.7e+05	1.7e+02 /4.6e+02	4.8e+03 /9.7e+03	1.5e+02 /4.2e+02	1.8e+06/ 4.3e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.7e+05 /3.8e+05	2.8e+02 /5.5e+02	2.0e+03 /7.2e+03	2.6e+02 /5.3e+02	1.8e+06/ 7.1e+05
deadBytes	F	2.2e+08 /2.4e+08	4.9e+08 /8.1e+08	3.5e+05 /4.0e+05	1.8e+02 /4.2e+02	3.5e+03 /6.9e+03	1.4e+02 /3.9e+02	1.8e+06/ 7.3e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.3e+05 /3.9e+05	1.6e+02 /4.0e+02	1.9e+03 /6.9e+03	1.3e+02 /3.6e+02	1.8e+06/ 7.3e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.3e+05/ 4.7e+05	3.7e+02/ 2.1e+02	4.8e+03 /8.7e+03	3.4e+02/ 1.9e+02	1.8e+06/ 4.1e+05
sweepUnmarked	M	2.7e+08 /7.8e+08	7.6e+08 /2.1e+09	1.9e+05 /2.5e+06	8.0e+04 /1.4e+05	4.0e+03 /1.7e+04	4.7e+03 /7.4e+03	1.8e+06/ 9.3e+05
totalHeapSize	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	1.1e+06/ 9.7e+05	5.6e+02 /7.7e+02	1.6e+04 /1.8e+04	3.8e+02 /5.9e+02	1.6e+06/ 3.8e+05
touchHotObjects	M	2.7e+08 /8.0e+08	7.7e+08 /2.2e+09	2.4e+05 /1.8e+06	8.7e+04 /1.1e+05	3.3e+03 /2.3e+04	4.5e+03 /5.8e+03	1.7e+06/ 9.3e+05
Total		2.1e+09/3.2e+09	4.9e+09/9.9e+09	3.8e+06/7.8e+06	1.7e+05/2.6e+05	4.5e+04/1.1e+05	1.1e+04/1.6e+04	1.6e+07/5.7e+06

Table 35: Per-pass performance for **PiecewiseFunctions**, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1255	0.1323	0.1445	0.4083	0.87 $\times$	1.05 $\times$	0.92 $\times$	0.32 $\times$
autorefineMaxLevel	F	4/8	50%	0.0209	0.0448	0.0193	0.2339	1.08 $\times$	2.14 $\times$	2.32 $\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0201	0.0456	0.0116	0.2332	1.73 $\times$	2.27 $\times$	3.92 $\times$	0.20 $\times$
diffPW	M	8/8	0%	0.1260	0.1321	0.1374	0.4035	0.92 $\times$	1.05 $\times$	0.96 $\times$	0.33 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0218	0.0471	0.0187	0.2370	1.17 $\times$	2.16 $\times$	2.52 $\times$	0.20 $\times$
norm2Estimate	F	5/8	38%	0.0258	0.0636	0.0326	0.1980	0.79 $\times$	2.47 $\times$	1.95 $\times$	0.32 $\times$
pmapCutHistogram	F	4/8	50%	0.0326	0.0555	0.0126	0.2412	2.58 $\times$	1.70 $\times$	4.40 $\times$	0.23 $\times$
truncateTolViolations	F	3/8	62%	0.0202	0.0462	0.0141	0.2001	1.43 $\times$	2.28 $\times$	3.27 $\times$	0.23 $\times$
Total				0.3930	0.5672	0.3908	2.155	1.01 $\times$	1.44 $\times$	1.45 $\times$	0.26 $\times$
Geomean								1.23 $\times$	1.80 $\times$	2.21 $\times$	0.25 $\times$

Table 36: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.6e+08	5.7e+08 /1.2e+09	1.3e+06 /2.0e+06	1.3e+05 /3.3e+05	1.3e+04 /2.9e+04	8.9e+03 /2.7e+04	4.5e+06/ 1.8e+06
autorefineMaxLevel	F	1.1e+08/ 9.8e+07	2.5e+08 /3.7e+08	2.5e+06/ 2.7e+05	8.7e+02/ 1.4e+02	7.1e+03/ 6.4e+03	8.6e+02/ 1.3e+02	6.0e+06/ 1.0e+06
compressMass	F	1.0e+08/ 5.9e+07	2.5e+08 /2.8e+08	2.2e+06/ 3.0e+04	7.9e+02/ 3.1e+02	2.0e+04/ 5.1e+03	7.8e+02/ 3.0e+02	6.2e+06/ 6.9e+05
diffPW	M	2.9e+08 /3.4e+08	5.9e+08 /1.3e+09	7.3e+05 /2.2e+06	1.5e+05 /2.4e+05	1.0e+04 /2.9e+04	1.2e+04 /2.9e+04	4.1e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.5e+07	3.3e+08 /4.9e+08	2.4e+06/ 5.0e+05	6.2e+02/ 2.5e+02	1.5e+04/ 9.4e+03	6.0e+02/ 2.4e+02	6.2e+06/ 1.5e+06
norm2Estimate	F	1.3e+08/ 8.8e+07	3.4e+08 /4.4e+08	1.5e+06/ 9.6e+05	8.2e+02/ 5.0e+02	1.3e+04 /1.8e+04	7.4e+02/ 4.0e+02	6.1e+06/ 9.8e+05
pmapCutHistogram	F	1.7e+08/ 6.4e+07	2.8e+08 /3.7e+08	1.7e+06/ 5.7e+04	1.1e+03/ 3.7e+02	1.0e+05/ 7.2e+03	1.0e+03/ 3.4e+02	4.7e+06/ 1.3e+06
truncateTolViolations	F	1.0e+08/ 5.8e+07	2.5e+08 /2.9e+08	2.1e+06/ 5.3e+04	6.7e+02/ 3.1e+02	1.8e+04/ 4.8e+03	6.4e+02/ 2.8e+02	6.3e+06/ 6.7e+05
Total		1.3e+09/1.2e+09	2.8e+09/4.7e+09	1.4e+07/6.0e+06	2.9e+05/5.7e+05	2.0e+05/1.1e+05	2.5e+04/5.7e+04	4.4e+07/9.8e+06

Table 37: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.7e+09	7.8e+08 /3.3e+09	2.8e+05 /5.1e+06	1.3e+05 /3.3e+05	8.4e+03 /4.1e+04	6.1e+03 /3.7e+04	4.1e+06/ 1.9e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	3.0e+05 /2.0e+06	2.9e+02 /1.5e+03	1.8e+03 /4.1e+04	2.6e+02 /6.7e+02	4.2e+06/ 7.1e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.2e+09	3.0e+05 /1.2e+06	4.0e+02 /2.0e+03	2.0e+03 /2.9e+04	3.7e+02 /4.5e+02	4.1e+06/ 4.1e+05
diffPW	M	3.4e+08 /1.7e+09	8.1e+08 /3.3e+09	2.9e+05 /5.2e+06	1.2e+05 /2.6e+05	5.9e+03 /4.6e+04	6.1e+03 /3.6e+04	4.1e+06/ 1.8e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.6e+05 /2.6e+06	3.4e+02 /4.4e+02	2.1e+03 /4.2e+04	3.2e+02 /4.1e+02	4.1e+06/ 1.0e+06
norm2Estimate	F	2.9e+08 /1.0e+09	5.1e+08 /1.3e+09	1.7e+06 /3.2e+06	4.6e+02 /1.4e+03	3.1e+04 /5.9e+04	3.6e+02 /1.1e+03	3.8e+06/ 9.2e+05
pmapCutHistogram	F	2.8e+08 /1.2e+09	5.0e+08 /1.3e+09	7.5e+05 /2.0e+06	3.5e+02 /2.2e+03	1.0e+04 /3.7e+04	3.2e+02 /5.7e+02	3.9e+06/ 7.2e+05
truncateTolViolations	F	2.3e+08 /1.0e+09	4.4e+08 /1.3e+09	3.1e+05 /1.3e+06	3.0e+02 /1.5e+03	3.5e+03 /2.7e+04	2.7e+02 /5.1e+02	4.1e+06/ 4.1e+05
Total		2.2e+09/1.0e+10	4.5e+09/1.4e+10	4.0e+06/2.3e+07	2.5e+05/6.0e+05	6.5e+04/3.2e+05	1.4e+04/7.7e+04	3.2e+07/7.9e+06

Table 38: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0833	0.1121	0.0890	0.3425	0.94 $\times$	1.35$\times$	1.26$\times$	0.33 $\times$
sum tree	F	5/5	0%	0.0293	0.0499	0.0270	0.2770	1.09 $\times$	1.70$\times$	1.85$\times$	0.18 $\times$
Total				0.1125	0.1619	0.1160	0.6195	0.97 $\times$	1.44$\times$	1.40$\times$	0.26 $\times$
Geomean								1.01 $\times$	1.51$\times$	1.53$\times$	0.24 $\times$

Table 39: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.2e+05 /1.2e+06	6.2e+04 /1.2e+05	3.7e+03 /1.4e+04	2.5e+03/ 2.3e+03	1.6e+06/ 1.1e+06
sum tree	F	1.5e+08/ 1.4e+08	4.6e+08 /5.4e+08	4.7e+04 /2.7e+05	1.2e+02/ 1.1e+02	7.0e+02 /8.9e+03	1.1e+02/ 9.6e+01	1.8e+06/ 1.2e+06
Total		4.2e+08/3.6e+08	1.2e+09/1.5e+09	4.7e+05/1.5e+06	6.2e+04/1.2e+05	4.4e+03/2.3e+04	2.6e+03/2.4e+03	3.4e+06/2.3e+06

Table 40: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.2e+08 /1.6e+09	9.5e+08 /2.4e+09	4.4e+05 /2.3e+06	6.7e+04 /1.8e+05	6.7e+03 /4.3e+04	3.2e+03 /5.0e+03	1.4e+06/ 9.7e+05
sum tree	F	2.5e+08 /1.4e+09	5.8e+08 /1.0e+09	6.9e+04 /2.5e+06	1.2e+02 /1.2e+03	1.3e+03 /4.3e+04	1.1e+02 /6.1e+02	1.5e+06/ 9.1e+05
Total		5.8e+08/3.0e+09	1.5e+09/3.4e+09	5.1e+05/4.7e+06	6.7e+04/1.8e+05	8.0e+03/8.6e+04	3.3e+03/5.7e+03	2.9e+06/1.9e+06

Table 41: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0158	0.0267	0.0101	0.1074	1.56$\times$	1.69$\times$	2.65$\times$	0.25 $\times$
countLazyNodes	F	4/10	60%	0.0137	0.0271	0.0065	0.1080	2.11$\times$	1.98$\times$	4.18$\times$	0.25 $\times$
countTerminals	F	3/10	70%	0.0132	0.0251	0.0096	0.0999	1.37$\times$	1.90$\times$	2.61$\times$	0.25 $\times$
decayTrieStats	M	10/10	0%	0.0760	0.0821	0.0860	0.2487	0.88 $\times$	1.08 $\times$	0.95 $\times$	0.33 $\times$
resetTraversalState	M	10/10	0%	0.0765	0.0834	0.0838	0.2323	0.91 $\times$	1.09 $\times$	0.99 $\times$	0.36 $\times$
sumPrefixFreq	F	3/10	70%	0.0135	0.0393	0.0114	0.1072	1.18$\times$	2.92$\times$	3.44$\times$	0.37 $\times$
sumSubtreeHints	F	3/10	70%	0.0131	0.0255	0.0080	0.1042	1.63$\times$	1.95$\times$	3.19$\times$	0.24 $\times$
Total				0.2218	0.3092	0.2155	1.008	1.03 $\times$	1.39$\times$	1.43$\times$	0.31 $\times$
Geomean								1.32$\times$	1.71$\times$	2.26$\times$	0.29 $\times$

Table 42: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	8.0e+07/ 4.4e+07	1.3e+08 /2.1e+08	1.5e+06/ 2.9e+05	1.5e+02/ 1.1e+02	5.3e+04/ 5.0e+03	1.5e+02/ 1.0e+02	3.7e+06/ 6.9e+05
countLazyNodes	F	6.9e+07/ 3.3e+07	1.3e+08 /1.6e+08	1.3e+06/ 2.1e+04	2.1e+02/ 1.8e+02	1.6e+04/ 3.3e+03	1.8e+02/ 1.5e+02	4.0e+06/ 6.5e+05
countTerminals	F	6.7e+07/ 2.8e+07	1.0e+08 /1.2e+08	1.3e+06/ 9.9e+03	2.8e+02/ 1.8e+02	2.6e+04/ 2.3e+03	2.7e+02/ 1.5e+02	3.9e+06/ 3.0e+05
decayTrieStats	M	1.7e+08 /2.1e+08	3.5e+08 /7.9e+08	7.7e+05 /1.3e+06	9.4e+04 /1.2e+05	5.9e+03 /1.9e+04	3.1e+03 /2.3e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.0e+08	2.8e+08 /7.0e+08	1.1e+06/ 1.0e+06	9.2e+04 /1.1e+05	7.9e+03 /1.3e+04	3.4e+03 /1.3e+04	3.3e+06/ 8.9e+05
sumPrefixFreq	F	6.8e+07/ 2.8e+07	1.1e+08 /1.2e+08	1.3e+06/ 1.2e+04	6.3e+02/ 4.1e+02	2.9e+04/ 2.6e+03	5.2e+02/ 2.9e+02	3.9e+06/ 2.8e+05
sumSubtreeHints	F	6.6e+07/ 2.8e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.3e+04	3.5e+02/ 1.6e+02	2.8e+04/ 2.4e+03	3.4e+02/ 1.5e+02	3.9e+06/ 3.2e+05
Total		6.8e+08/5.8e+08	1.2e+09/2.2e+09	8.7e+06/2.7e+06	1.9e+05/2.2e+05	1.7e+05/4.8e+04	8.0e+03/3.6e+04	2.6e+07/4.3e+06

Table 43: Per-pass PAPI counters for **Trie** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.5e+08	2.3e+08 /7.5e+08	1.1e+06 /1.7e+06	2.3e+02 /1.1e+03	2.9e+03 /3.2e+04	2.2e+02 /8.6e+02	2.9e+06/ 4.6e+05
countLazyNodes	F	1.4e+08 /5.5e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.1e+06	1.2e+02 /8.1e+02	2.6e+03 /2.2e+04	1.2e+02 /3.7e+02	2.8e+06/ 3.6e+05
countTerminals	F	1.3e+08 /5.1e+08	2.0e+08 /7.3e+08	1.2e+06/ 6.9e+05	1.4e+02 /1.1e+03	1.5e+03 /1.7e+04	1.2e+02 /2.6e+02	2.9e+06/ 1.9e+05
decayTrieStats	M	1.9e+08 /1.0e+09	4.5e+08 /2.1e+09	3.7e+05 /5.3e+06	9.0e+04 /3.7e+05	4.4e+03 /3.9e+04	3.5e+03 /3.7e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	2.0e+08 /9.6e+08	3.7e+08 /2.0e+09	3.2e+05 /3.2e+06	1.1e+05 /1.6e+05	5.8e+03 /2.9e+04	3.9e+03 /3.9e+04	3.0e+06/ 9.7e+05
sumPrefixFreq	F	1.2e+08 /5.3e+08	2.1e+08 /7.3e+08	1.5e+06/ 6.9e+05	5.3e+02 /1.4e+03	1.3e+03 /1.7e+04	4.3e+02/ 4.0e+02	2.5e+06/ 1.9e+05
sumSubtreeHints	F	1.3e+08 /5.3e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.9e+05	1.3e+02 /9.5e+02	1.3e+03 /1.7e+04	1.2e+02 /2.5e+02	2.9e+06/ 1.9e+05
Total		1.0e+09/4.6e+09	1.9e+09/7.8e+09	7.9e+06/1.3e+07	2.0e+05/5.4e+05	2.0e+04/1.7e+05	8.5e+03/7.8e+04	2.0e+07/3.6e+06